

A Detailed Thermal Resistance Network Analysis of FCBGA Package

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Abstract: Using thermal models to describe the heat dissipation process of FCBGA is a significant topic in the field of packaging. However, the thermal resistance model considering the structure of each part of the chip is still ambiguous and rare, but it is quite desirable in engineering. In this work, we propose a detailed thermal resistance network model, and describe it by using thermal conduction resistance and thermal spreading resistance. For a striking FCBGA case, we calculated the thermal resistance of each part of the structure according to the temperature field simulated by COMSOL. The thermal resistance network can be used to predict the temperatures in the chip under different conditions. For example, when the power changes by 40%, the relative error of junction temperature prediction is only 0.24%. The function of the detailed thermal resistance network in evaluating the optimization space and determining the optimization direction is clarified. This work illustrates a potential thermal resistance analysis method for electronic devices such as FCBGA.

Keywords: FCBGA package; thermal resistance network; thermal spreading resistance; junction temperature

1. Introduction

The development of electronic packaging to high integration and miniaturization increases the risk of thermal failure caused by high junction temperature [1–3], which can be effectively addressed through thermal analysis and thermal design [4, 5]. The problem is particularly critical for 5G (the 5th Generation Mobile Communication Technology): generally, due to the rapid increase of signal transmission capacity of 5G base station [6, 7], the power consumption of 5G equipment is four times that of 4G equipment [8, 9], which means a greater heat dissipation challenge. Flip chip ball grid array (FCBGA) chip is the key of 5G base station and has been widely concerned at present [10–14]. Its

junction temperature extremely affects the performance and service time of the 5G equipment. Therefore, it is of great significance to study the weak point of heat dissipation through thermal analysis, so as to improve the heat dissipation capacity and ensure the safe operation of the chip.

The specific form of thermal analysis is to use thermal resistance model/network to describe the heat dissipation path of the chip. At present, the two-resistances model [15], the star network thermal resistance model [16] and the boundary-conditions-independent thermal resistance network model [17, 18] are widely studied. By simplifying the heat dissipation path, these models can be used to predict the junction temperature of the chip. However, these models studied the chip under

unpacked state, excluding the packaging conditions such as a large heat sink in the real equipment. The heat dissipation is simulated by giving the convective heat transfer coefficient on the upper surface of the metal case (lid), which ignores the heat dissipation difference at different positions of the surface. In addition, only the total thermal resistance of each path is given, but the detailed composition of each path has not been explored. However, the chip designers need the size and proportion of thermal resistance of each part in the path, which is important for understanding the thermal resistance optimization space and determining the optimization way.

The chip package can be regarded as a stack of plates with different cross-sectional areas. However, its total thermal resistance (R_{total}) is not equal to the sum of the thermal conduction resistance (R_{ID}) of each plate. This is because the thermal spreading process on the plate surface needs to be considered, which will lead to temperature differences at different locations. The thermal resistance of a flat plate with a small area heat source has been studied in the previous literature [19, 20]. By derivation, R_{total} is separated into R_{ID} and thermal spreading resistance (R_S), and the series relationship between them is pointed out [19]. R_S exists in two forms in the thermal resistance network of FCBGA chip. The first is caused by the uneven heat flux distribution of the heat source, and the other is caused by the sudden change of cross-sectional area between different plates.

In this study, the FCBGA package model is constructed. The characteristic details such as uneven heat flux distribution at the heat source and multi-layer micro structure are retained. Then, the finite element simulation is carried out by using COMSOL Multiphysics. According to the simulated temperature field, we give the calculation method of R_{ID} and R_S in detail and the thermal resistance network. Furthermore, the role of thermal resistance network in temperature prediction (junction temperature, case temperature and

other temperatures) and thermal resistance optimization is discussed.

2. Methods/Research Model

2.1 Model setup

A three dimensional computational model of FCBGA package with flat lid is developed to study its thermal resistance network. The y -axis (front) and z -axis (top) views of the actual physical model are shown in Fig. 1(a) and Fig. 1(b). The model includes heat sink, chip, PCB (printed circuit board) and air channel, which are marked in red font. The air channel is adopted to simulate the process that heat dissipated by natural convection at heat sink. According to the actual installation mode of the package [21], gravity ($g=9.8 \text{ m/s}^2$) is set in the direction parallel to the fins (also y -axis) in simulation. Due to the existence of temperature difference and gravity in the calculation domain, the cooling air will enter from the lower open boundary along the y -axis direction and leave the domain from the upper open boundary. The size of each part of the model varies greatly, so we use schematic diagram Fig. 1(c) to show the structure more clearly. The chip consists of lid, TIM1 (thermal interface materials), die, bump, substrate and solder balls, which are marked in black font. Bump and solder ball are modeled as flat plates rather than ball arrays to improve computational efficiency [22–24]. See supplementary material S3 for more details. Please note that the dimensions in the schematic diagram are not strictly accurate and are only used to illustrate the structure. The heat source of chip is on the bottom of the die, with a total of 23.09 W. The heat will be dissipated along the upward and downward paths, which are called path 1 and path 2 respectively. The purpose of this study is to clarify the composition and calculation method of the thermal resistance of each part in the two paths. The detailed heat flux distribution and values are given in supplementary materials (Fig. S1 and Table S1).

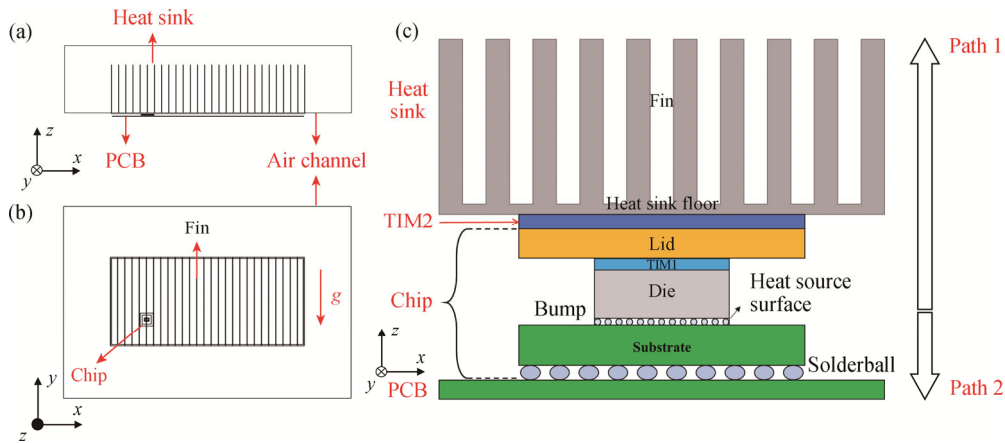


Fig. 1 (a) Front view of the computational model; (b) top view of the computational model; (c) schematic diagram of the flip chip

The detailed dimensions and physical parameters are shown in Table 1. The thermal conductivity of bump, solder ball, PCB and substrate is anisotropic [22–25], so we use the form of $\{k_x, k_y, k_z\}$ to describe their details. As for the position of the chip, we locate it by the distance between the lower left corner of the PCB and the lower left corner of the chip (60 mm and 40 mm in the x and the y direction in Fig. 1(b)).

2.2 Boundary conditions

The boundary conditions are as follows:

- (1) The heat flux distribution at the bottom of the die is given. (see Section 2.1 for details)
- (2) Uniform convective heat transfer coefficient is applied to the bottom of PCB. ($h=5 \text{ W}/(\text{m}^2 \cdot \text{K})$)
- (3) The air domain is included and a non-isothermal flow model is adopted to calculate the heat loss at the heat sink. (ambient temperature $T_0=328.15 \text{ K}$)
- (4) The effect of radiation is ignored [26].
- (5) All other non-contact surfaces are adiabatic.

2.3 Numerical method

In this study, COMSOL Multiphysics is employed to establish the model and perform the finite element simulation. The unstructured grids of tetrahedral elements are generated in the studied chip system, as shown in Fig. 2(a), and its perspective is consistent with Fig. 1(b). In addition, for some local parts of the model (such as die, TIM1, lid and substrate, etc.), we first construct a free triangular mesh in the x - y plane (Fig. 2(b)), and generate a three-dimensional mesh by sweeping to improve grid quality (Fig. 2(c)).

In order to verify the reasonableness of the grid, we check the junction temperature T_j (also called maximum temperature) of the chip. Fig. 3(a) shows that when the grid number is 1.3×10^6 and 2.2×10^6 , T_j is nearly constant, and the relative error is only 0.012%. Thus, the grid number is chosen as 1.3×10^6 in this study. In addition, the height $H_{\text{Air channel}}$ along z -axis of the air channel will also affect the convective heat transfer process. Therefore, we study the influence of $H_{\text{Air channel}}$ on T_j . As shown in

Table 1 Detailed dimensions and physical parameters of the model

	Size $x/\mu\text{m}$	Size $y/\mu\text{m}$	Size $z/\mu\text{m}$	$k/\text{W} \cdot (\text{m} \cdot \text{K})^{-1}$
Fins (28)	800	185 000	100 000	130
Heat Sink Floor	405 000	185 000	2000	130
TIM2	27 000	27 000	600	6
Lid	27 000	27 000	1000	385
TIM1	10 160	7300	67	2.2
Die	10 160	7300	790	117
Solder ball	27 000	27 000	500	0.1, 0.1, 35
Bump	10 160	7300	60	0.5, 0.5, 33.86
PCB	400 000	180 000	1800	95.76, 95.76, 0.52
Substrate	27 000	27 000	1488	0.48, 0.48, 16.75
Air channel	600 000	400 000	1 400 000	—

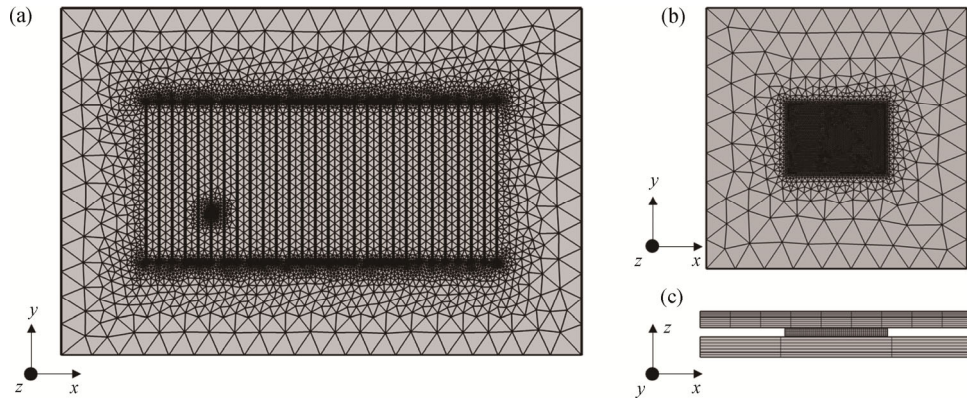


Fig. 2 Mesh configuration in FCBGA package: (a) top view of the whole domain; (b) top view (x - y plane) of chip; (c) front view of chip

Fig. 3(b), when the height is 140 000 μm or 150 000 μm , the relative error is only 0.0065%. That is, it is reasonable to use an air channel with a height of 140 000 μm . By adopting the above meshing method and grid number, the accuracy and time-consuming of calculation are balanced.

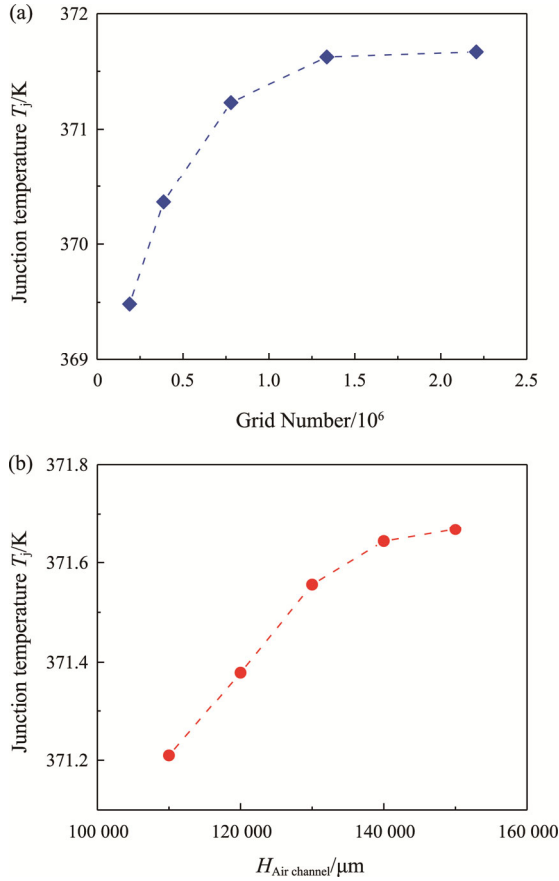


Fig. 3 (a) Grid independence test of the FCBGA package, (b) air channel height independence test

3. Results and Discussions

In this study, the model is mainly used to analyze the thermal resistance network. Firstly, the temperature distribution contours are obtained through numerical calculation, and the thermal conduction resistances (R_{1D}) of each part are calculated. And then the thermal spreading resistance (R_S) is discussed in detail, including its source and calculation method. Then the value and proportion of each part of thermal resistance are given in detail. The calculation method and results are meaningful in engineering applications, such as rapid junction temperature prediction and pointing out the direction of thermal resistance optimization.

3.1 Basic simulation results

Fig. 4 shows the temperature distribution of FCBGA package obtained in this study. In Fig. 4(a), the chip is located in the brighter position (meaning high temperature). Due to the buoyancy force caused by the temperature difference, the natural convection of air will occur. The black curve lines with arrow are used to indicate the air flow path. Fig. 4(b) and 4(c) show the temperature distribution at the die location, and they use the same colorbar. It can be observed that the temperature of the heat source surface on the die is uneven, even if the die material is silicon with high thermal conductivity (117 W/(m·K)). Fig. 4(c) shows the die and some of its upper and lower structures, belonging to heat dissipation path 1 and path 2, respectively. It is obvious that the temperature on the lower side is higher than that on the upper side, which indicates that the thermal resistance of the path 2 is greater, which will be analyzed in detail and the specific proportion will be given later.

For more details, some maximum and average temperatures of each surface required for subsequent calculations are given in Table 2. The subscripts b and t

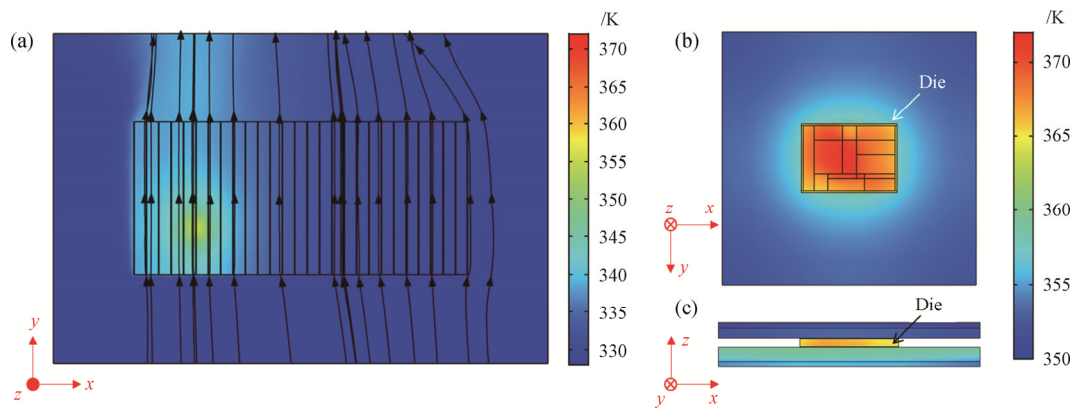


Fig. 4 (a) Temperature distribution contours of the system; (b) temperature distribution contours at the die (bottom view); (c) temperature distribution contours at the die (front view)

of the temperature in the table represent the bottom and top surfaces of the structure (in the z -axis direction). The subscripts ave or max indicates that this temperature is the average or maximum temperature of the surface. For example, $T_{\text{Die,b,max}}$ represents the maximum temperature at the bottom surface of the die.

Table 2 Temperatures of each surface

	T/K		T/K
$T_j(T_{\text{Die,b,max}})$	371.62	$T_{\text{HSF,t,ave}}$	334.25
$T_{\text{Die,b,ave}}$	368.42	$T_{\text{Bump,b,ave}}$	368.30
$T_{\text{Die,t,ave}}$	366.75	$T_{\text{Substrate,b,ave}}$	357.41
$T_{\text{TIM1,t,ave}}$	359.22	$T_{\text{Solderball,b,ave}}$	357.32
$T_{\text{Lid,t,ave}}$	354.33	$T_{\text{PCB,b,ave}}$	341.36
$T_{\text{TIM2,b,ave}}$	351.82	T_0	328.15

3.2 Thermal resistance network

The thermal resistance of FCBGA package including thermal conduction resistance R_{ID} and thermal spreading resistance R_{S} . The following will begin with the introduction of R_{ID} to clarify the source and calculation method of R_{S} . From the perspective of thermal analysis, the model studied in this work can be regarded as the stacking of plates (except fins and air channel). In general, the calculation method of R_{ID} of one-dimensional heat transfer of plate is as follows:

$$R_{\text{ID}} = \frac{l}{kA} \quad (1)$$

where R_{ID} is the thermal conduction resistance. k is the thermal conductivity. A and l are the area and thickness of plate respectively.

On the other hand, R_{ID} can also be calculated by using the simulated temperature field:

$$R_{\text{ID}} = \frac{T_{\text{b,ave}} - T_{\text{t,ave}}}{Q_i} \quad (2)$$

where $T_{\text{b,ave}}$ and $T_{\text{t,ave}}$ are the average temperature of the bottom and top surfaces of the plate, respectively. Q_i is the heat flow through the section. Moreover, it can be proved that it is equivalent to Eq. (1) (see supplementary materials for derivation). In addition, the above relationship is satisfied even if the thermal conductivity of materials is anisotropic (not shown here). Thus, the R_{ID} of most structures in the package can be calculated.

Structures with anisotropic thermal conductivity in chips include solder ball, bump, PCB and substrate, all of which have thermal conductivity of the form $\{k_x, k_y, k_z\}$. By substituting k_z (thermal conductivity along the heat transfer path direction) into Eq. (3), we can derive R_{ID} . Similarly, using the average temperature ($T_{\text{b,ave}}$ and $T_{\text{t,ave}}$) of the bottom and top surfaces of the structure, we can derive another R_{ID} . See Table 3 for comparison.

Table 3 Comparison of R_{ID} and R_{ID}'

	$\{k_x, k_y, k_z\}/\text{W} \cdot (\text{m} \cdot \text{K})^{-1}$	$R_{\text{ID}}/\text{K} \cdot \text{W}^{-1}$	$R_{\text{ID}}'/\text{K} \cdot \text{W}^{-1}$
Solder ball	{0.1, 0.1, 35}	0.0196	0.0196
Bump	{0.5, 0.5, 33.86}	0.0239	0.0239
PCB	{95.76, 95.76, 0.52}	0.0481	0.0481
Substrate	{0.48, 0.48, 16.75}	0.1219	0.1219

The result shows that even for anisotropic materials, the one-dimensional thermal conduction resistance still depends on the thermal conductivity along the direction of the heat transfer path. However, it should be noted that the in-plane thermal conductivity $\{k_x, k_y\}$ also affect the total thermal resistance of the system, but do not directly change the thermal conduction resistance of the structure. For example, the sudden change of cross-sectional area (SCCSA) between PCB and solder ball will cause thermal spreading resistance. Although the thermal conduction resistance value R_{ID} of PCB is determined, the anisotropic thermal conductivity of the solder ball and PCB will affect the heat diffusion at SCCSA. Lower k_x and k_y will make it harder for the heat to diffuse at the variable cross section, resulting in a larger $(T_{\text{Solderball,b,ave}} - T_{\text{PCB,b,ave}})$. It will change the thermal spreading resistance $R_{\text{S,PCB}}$. To summarize, the in-plane thermal conductivity of anisotropic structures ultimately affects the thermal spreading resistance R_{S} induced at the area mutation. Therefore, the equations mentioned above are also applicable to anisotropic materials. Conversely, if a certain area mutation is found to have a large R_{S} , it can be improved by increasing the in-plane thermal conductivity of the large-area structure (e.g., PCB) or the small-area structure (e.g., solder ball).

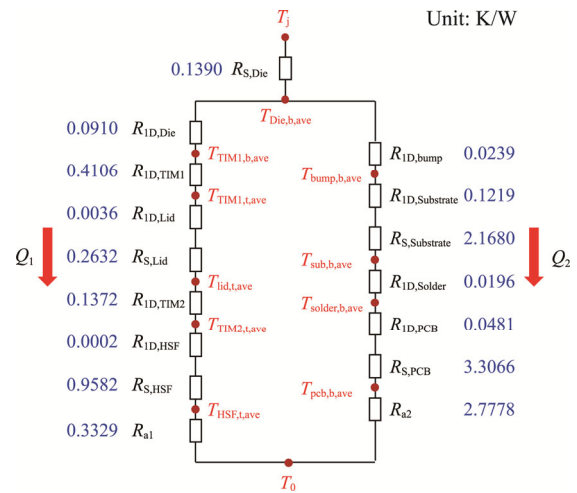


Fig. 5 Thermal resistance network of FCBGA package

In order to explain the thermal resistance of FCBGA package more clearly, the thermal resistance network

diagram shown in Fig. 5 is given. The thermal resistance values are indicated in Fig. 5 in blue font. Heat is transferred to the surrounding through path 1 and path 2 (Q_1 and Q_2 respectively). All the heat lost by the chip through path 1 will pass through TIM1. Therefore, according to the equivalence relationship between Eqs. (1) and (2), we can get $Q_1=18.33$ W. Similarly, $Q_2=4.76$ W can be obtained. Other structures such as die are used for verification, and the results are consistent (not shown there). The thermal resistances of path 1 and path 2 are in parallel, and each thermal resistance in each path is in series.

There are some R_S in the network, because the heat spreading in the x - y plane needs to be considered. Firstly, the uneven heat flux distribution at the heat source will introduce the thermal spreading resistance. Due to the uneven heat flux distribution of the heat source (the bottom of the die), the surface temperature is uneven. The maximum (junction) temperature and the average temperature of the surface can both be obtained in the simulation to define thermal resistance. But in reality, engineers pay more attention to the maximum temperature to judge the risk of thermal failure. In order to include the information of the temperature difference $\Delta T=T_{\max}-T_{\text{ave}}$ of the heat source surface in the thermal resistance network, R_S is introduced. On the other hand, this temperature difference is due to the limitation of heat spreading ability on the x - y plane of the die. If $k_{\text{die}}=+\infty$ W/(m·K), the temperature unevenness of the heating surface will be eliminated, and R_S will be equal to 0 K/W. Therefore, it is classified as thermal spreading resistance. In most FCBGA packages (including the case in this study), the heat flux distribution is nonuniform, so the in-plane temperature is bound to be uneven. In other words, the source of R_S here is the uneven heat flux distribution of the chip. The significance of R_S is to describe the difference between the junction temperature and the average temperature of the heat source. In summary, the calculation approach of $R_{S,\text{die}}$ is:

$$R_{S,\text{die}} = \frac{(T_j - T_{\text{die,b,ave}})}{Q} \quad (3)$$

$$R_{\text{total}} = R_{S,\text{die}} + \left(\frac{1}{R_{1D,\text{die}} + R_{1D,\text{TIM1}} + R_{1D,\text{lid}} + R_{S,\text{lid}} + R_{1D,\text{TIM2}} + R_{1D,\text{HSF}} + R_{S,\text{HSF}} + R_{a1}} + \frac{1}{R_{1D,\text{bump}} + R_{1D,\text{sub}} + R_{S,\text{sub}} + R_{1D,\text{solder}} + R_{1D,\text{pcb}} + R_{S,\text{pcb}} + R_{a2}} \right)^{-1} \quad (7)$$

The thermal resistance network can be used for rapid temperature prediction when the heat transfer conditions such as power and external environmental change. The temperature distribution contours of the heat source surface under different Q_{total} are shown in Fig. 6. Figs.

Secondly, the sudden change of area will also lead to the thermal spreading resistance (such as the contact between TIM1 and lid). Heat is transferred to the bottom of lid through TIM1, and the heat transfer cross-sectional area becomes larger. However, the temperature of the lid is also uneven in the x - y plane, and the surrounding temperature is lower than the central temperature. This non-uniformity of temperature makes a large part heat transfer capacity of lid wasted. This phenomenon is especially obvious at the contact between TIM2 and the heat sink floor (HSF), because the huge area mutation makes temperature at many locations of the heat sink close to surrounding temperature (as shown in Fig. 4(a)), which means that many locations hardly play the role of heat dissipation. At this time, the thermal resistance of the lid or the heat sink floor will be greater than the R_{1D} , and the excess part is R_S . According to the simulation results, the temperatures at different locations can be obtained, and the thermal resistance can be calculated by using the temperature and heat flow. For example, at the contact of TIM1 and lid, we can calculate $R_{S,\text{lid}}$ using the following equation:

$$R_{S,\text{lid}} = R_{\text{TIM1-lid}} - R_{1D,\text{lid}} = \frac{T_{\text{TIM1,t,ave}} - T_{\text{lid,t,ave}}}{Q_1} - R_{1D,\text{lid}} \quad (4)$$

where $R_{\text{TIM1-lid}}$ is the total thermal resistance between the top surface of TIM1 and top surface of the lid.

In the thermal resistance network, R_{a2} represents the convective thermal resistance between the bottom of PCB and the surrounding:

$$R_{a2} = 1/hA \quad (5)$$

The total thermal resistance of the system is:

$$R_{\text{total}} = \frac{T_j - T_0}{Q} \quad (6)$$

R_{a1} represents the thermal resistance from the fins to the air. Since the air channel is adopted, the thermal resistance of this part cannot be calculated directly. However, in the above, the total thermal resistance and other thermal resistances have been calculated, so R_{a1} can be calculated according to the following equation:

6(a)–(e) are planar plots and Figs. 6(f)–(j) are deformed plots corresponding to $Q_{\text{total}}=0.6Q_0, 0.8Q_0, Q_0, 1.2Q_0, 1.4Q_0$, respectively. It can be found that the five plots (a)–(e) are basically identical, which indicates that even if Q_{total} changes, the heat dissipation characteristics of the

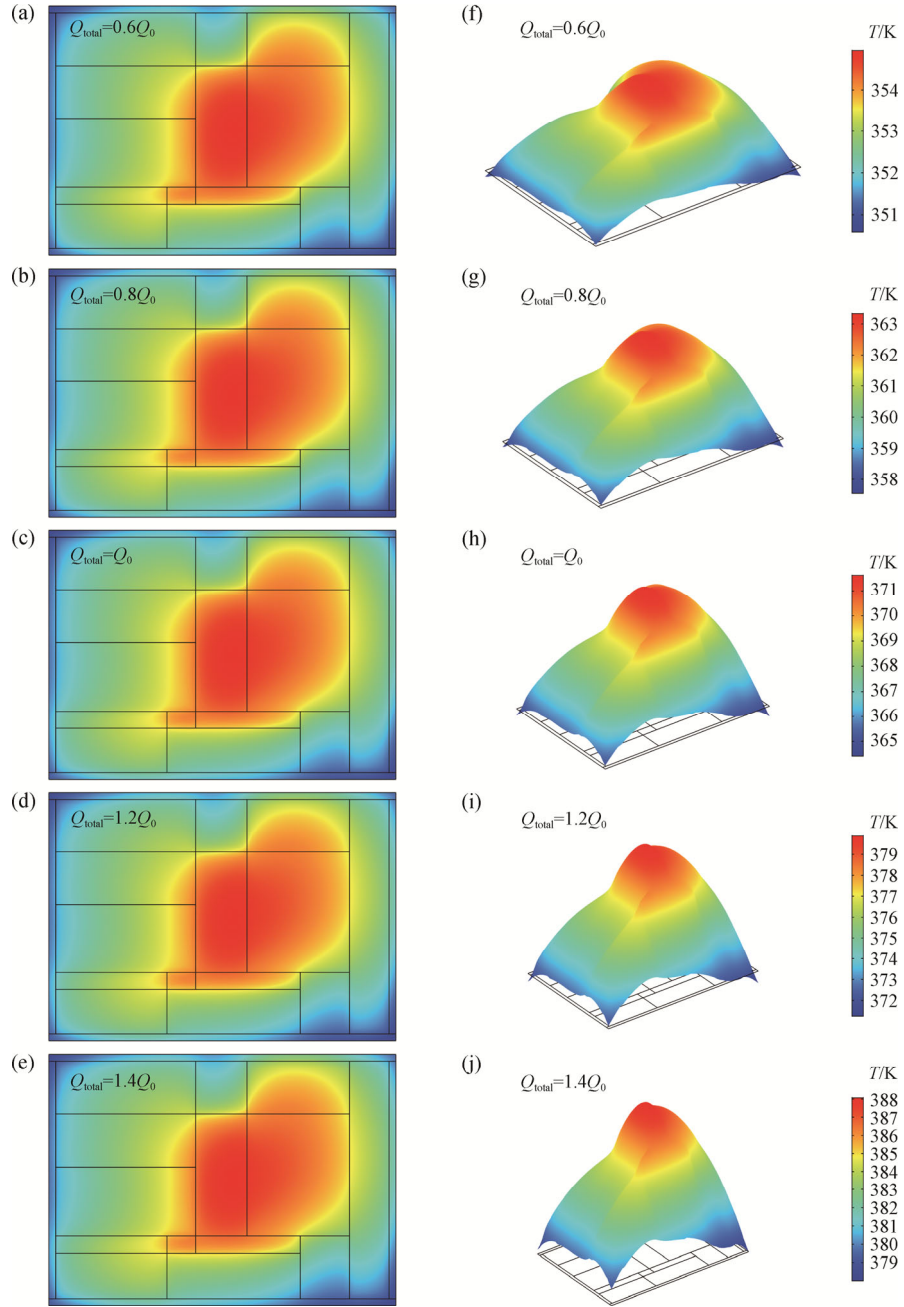


Fig. 6 (a)–(e) Temperature distribution contours of the heat source surface under different Q_{total} ; (f)–(j) temperature distribution contours of the heat source surface with deformation under different Q_{total}

chip do not change with it, indirectly illustrating the feasibility of thermal resistance network plots for temperature prediction under variable power. However, they have different legends, especially for the extreme temperature difference. Therefore, we plotted the plots (f)–(j) in which the magnitude of the temperature difference are more visualized. With the same overall temperature distribution trend, the degree of deformation in z -direction gradually increases with the increase of Q_{total} , which means that the difference between the

maximum in-plane temperature (junction temperature) and its average temperature increases. However, it is reasonable that the ratio of ΔT to Q is $R_{S,\text{die}}$, and the current temperature difference variation precisely illustrates the stability of the $R_{S,\text{die}}$ in the proposed thermal resistance model.

Fig. 7 shows the comparison between the temperature simulated by COMSOL and the temperature calculated according to thermal resistance network under different Q_{total} . In Fig. 7(a), the hollow diamond and circle markers

represent the maximum temperature of the heat source surface, and the solid markers represent the average temperature of the upper surface of the lid. The temperature of these two locations is of great concern to engineers. We find a good agreement between the COMSOL results and the prediction results. It can also be seen from the figure that at the point far away from $Q_{\text{total}}=23.09$ W, the difference between the predicted temperature and the simulated temperature begins to become larger. This is because the thermal spreading resistance R_S in the thermal resistance network will change with the external conditions. When the external conditions change, the utilization of the heat transfer capacity of the system will change, so the value of R_S will also change. To illustrate this point, use the change of $R_{S,\text{die}}$ with Q_{total} as an example (as shown in Fig. 7(b)). It can be seen that $R_{S,\text{die}}$ has an downward trend with the increase of Q_{total} . It is understandable that when the external heat dissipation environment is unchanged, the increase of Q_{total} means greater heat dissipation pressure, which will cause the temperature of the heat source surface to be higher and more uniform, so that the R_S will change accordingly. However, the change of $R_{S,\text{die}}$ is very small, so the maximum relative error in junction temperature prediction is only 0.24%, which is

accurate enough and acceptable.

The thermal resistance network can guide the optimization of the chip. Firstly, $R_{S,\text{die}}$ is a relatively large value, which means that chip designers can reduce it by optimizing heat flux layout. Another work of ours can illustrate the model's function. In this study, we found that the $R_{S,\text{die}}$ of FCBGA chips is large and can be attributed to the heat source layout, so in another study we carried out the optimization of the heat source layout and obtained a 30.94% reduction in the thermal spreading resistance $R_{S,\text{die}}$ and a 2.9°C reduction in the junction temperature by using bionic method. It can be considered as an application case guided by this model [27].

In addition, the total thermal resistances of path 1 and path 2 are 2.1969 K/W and 8.4658 K/W respectively. It can be seen that the latter is much greater than the former. Therefore, path 1 can be considered as the main heat dissipation path, which is also consistent with the real situation. Fig. 8 shows the proportion of each thermal resistance on path 1, which can guide the thermal resistance optimization. For example, the $R_{S,\text{HSF}}$ accounts for a large proportion. Engineers can introduce vapor Chamber to improve the heat dissipation efficiency [28–30], which will greatly reduce $R_{S,\text{die}}$. The calculation results are shown in Table 4. $R_{S,\text{HSF}}$ decreases by

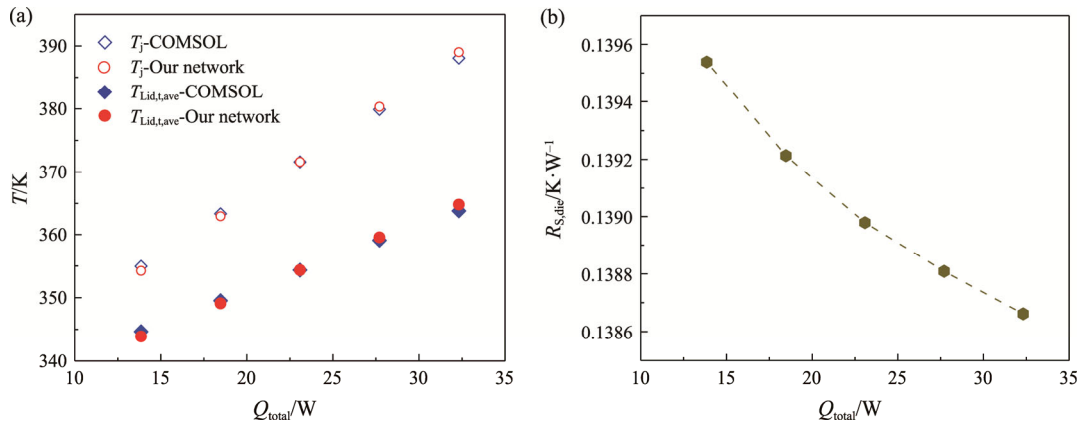


Fig. 7 (a) Comparison between the temperature simulated by COMSOL and the temperature calculated according to thermal resistance network under different Q_{total} ; (b) change of $R_{S,\text{die}}$ with Q_{total}

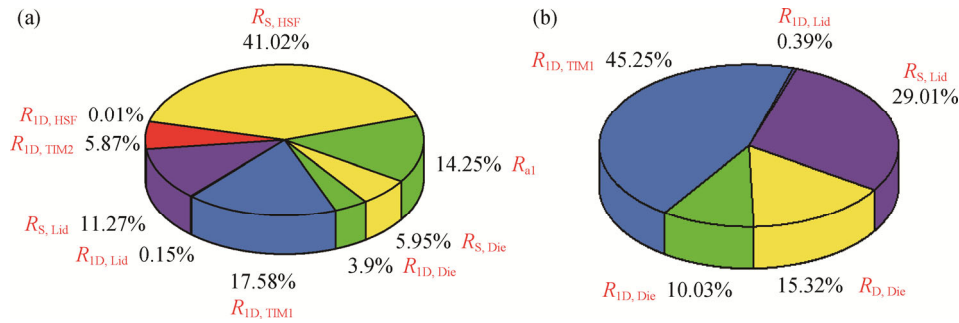


Fig. 8 (a) Proportional pie chart of thermal resistance of path 1; (b) Pie chart of internal thermal resistance of the chip along path 1

86.7% and the junction temperature decreases by 13.96 K. The temperature distribution on the top surface of heat sink floor is shown in Fig. 9.

Table 4 Comparison of $R_{S,HSF}$ and T_j

HSF	$k/W \cdot (m \cdot K)^{-1}$	$R_{S,HSF}/K \cdot W^{-1}$	T_j/K
Al	130	0.9582	371.62
Heat pipe/VC	2000	0.1270	357.66

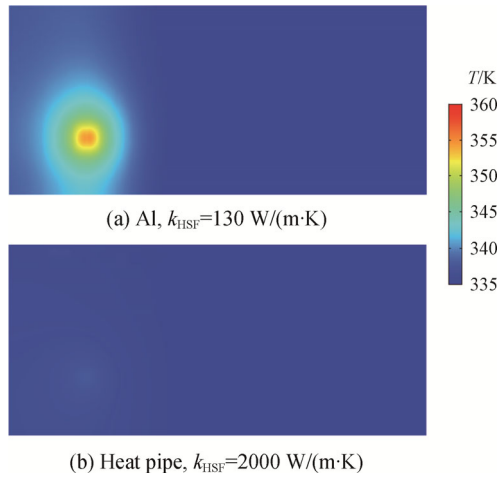


Fig. 9 Comparison of the top surface temperature of heat sink floor before and after using heat pipe/VC

Furthermore, according to the thermal resistance network, $R_{ID,TIM1}$ is the second largest thermal resistance in path 1. Although TIM1 is only 67 μm thick and seems insignificant, replacing it with a higher quality thermal interface material can reduce the thermal resistance and junction temperature greatly.

Based on the above analysis, we illustrate the detailed thermal resistance network of FCBGA package. In brief, on the premise that path 1 and path 2 are used to describe the heat dissipation, each path is described in detail by using R_{ID} and R_S . In particular, the calculation methods of R_S caused by uneven heat flux distribution and sudden change of heat transfer area are discussed. In addition, heat is mainly lost through path 1 (from the chip to the heat sink). In this study, the thermal resistance of path 2 is about four times that of path 1. Furthermore, the two main uses of the thermal resistance network model are described: (1) Temperature prediction: it can be used to predict the junction temperature, case temperature and other temperatures under different heating conditions. The relative error between the COMSOL results and the prediction results is only 0.24%. (2) Optimization space analysis: it is pointed out that $R_{S,die}$ is large, and the thermal resistance of this part should be reduced by optimizing the heat flux distribution. And the proportion of thermal resistance of each part in the path 1 is given to

help designers better understand the optimization space and determine the optimization way. The above analysis method can not only be applied to FCBGA package, but also has potential application value in light-emitting diode (LED) [31], insulated gate bipolar transistor (IGBT) [32] and other electronic devices.

4. Conclusions

The model of FCBGA package with flat lid and actual size heat sink is established and studied. The temperature field is calculated by COMSOL, and the detailed thermal resistance network composed of thermal conduction resistance R_{ID} and thermal spreading resistance R_S is given. The thermal resistance network model can be used to quickly predict the junction temperature, case temperature and other temperatures in the system. When the total power change is $\pm 40\%$, the relative error of predicting the junction temperature is only $\pm 0.24\%$, which has high enough accuracy. In addition, the thermal resistance of each part can be seen intuitively according to the network, especially R_S , which can be used as important reference information to assist engineers in thermal design and thermal resistance optimization.

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Conflict of Interest

MA Weigang is an editorial board member for Journal of Thermal Science and was not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

Electronic Supplementary Materials

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